

1D Helmholtz EM Solver — Design, Validation, and Applications

APC 523 Final Project

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Zhiping Li conceived the idea, performed the theoretical derivation, and debugged the code. *Xuyang Xu* wrote the report. The code was developed with the assistance of AI.

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1. Problem and Solver Overview

We solve the 1D Helmholtz equation for monochromatic EM plane-wave scattering at a dielectric transition layer. The computational domain is divided into three zones:

Zone	Range	Description
Region 1	$z \leq z_1$	Semi-infinite homogeneous medium, ε_1
Transition	$z_1 \leq z \leq z_2$	Inhomogeneous medium, $\varepsilon_r(z)$ varies continuously
Region 2	$z \geq z_2$	Semi-infinite homogeneous medium, ε_2

Sign convention: time factor $e^{-j\omega t}$; right-going wave $\propto e^{-jk_z z}$. The transverse wavenumber $k_x = k_0 \sqrt{\varepsilon_1} \sin \theta$ is conserved.

Governing equations

TE (S) polarisation — E_y satisfies the standard Helmholtz equation:

$$\frac{d^2 E_y}{dz^2} + [k_0^2 \varepsilon_r(z) - k_x^2] E_y = 0$$

TM (P) polarisation — H_y satisfies a modified equation with a drift term:

$$\frac{d^2 H_y}{dz^2} - \frac{1}{\varepsilon_r} \frac{d\varepsilon_r}{dz} \frac{dH_y}{dz} + [k_0^2 \varepsilon_r(z) - k_x^2] H_y = 0$$

Two independent numerical methods

Method	Principle	JAX back-end	Order
ODE shooting	Backward IVP from z_2 to z_1 via diffrax	JIT + GPU	5th (Dopri5)
FEM	Galerkin weak form, linear/quadratic elements, Robin BCs	JAX dense solver	2nd (P1) / 4th (P2)

Both methods are implemented in `main.py` and exposed through the single class `HelmholtzSolver1D`.

ODE solvers available

Solver	Family	Order	Stages	Adaptive
Heun	Explicit RK	2	2	yes (embedded)
Tsit5	Explicit RK (FSAL)	5	7	yes
Dopri5	Explicit RK (FSAL)	5	7	yes
KenCarp4	IMEX ARK	4	6	yes
SemiImplicitEuler	Symplectic	1	2	no

The Butcher tableaux for the explicit RK methods are shown below.

Fixed-step sweep (`fixed_step=True`): each solver uses exactly `n_trans=res_trans` steps. This reveals the theoretical order: log-log slope $\approx -p$ in error vs `res_trans`.

Note: KenCarp4 is IMEX — `fixed_step=True` is overridden; its order-4 convergence is still visible. `SemiImplicitEuler` has no error estimator and is always fixed-step (order 1).

Adaptive sweep (`fixed_step=False`): `res_trans` is only a step-size hint; PID controller adapts. Dopri5 and Tsit5 reach machine precision; Heun saturates earlier.

2. Basic Usage

Instantiation

```
from main import HelmholtzSolver1D
```

```
# Define the permittivity profile (any Python callable  $z \rightarrow \text{complex}$ )
```

Butcher Tableaux (values from diffrax)

Heun (Explicit RK2, order 2)

c \ k	k1	k2
0.000	0	0
1.000	1.0000	0
b	0.5000	0.5000
b*	1.0000	0

Tsit5 (Explicit RK5, FSAL, order 5)

c \ k	k1	k2	k3	k4	k5	k6	k7
0.000	0	0	0	0	0	0	0
0.161	0.1610	0	0	0	0	0	0
0.327	-0.0085	0.3355	0	0	0	0	0
0.900	2.8972	-6.3594	4.3623	0	0	0	0
0.980	5.3259	-11.7489	7.4955	-0.0925	0	0	0
1.000	5.8615	-12.9210	8.1594	-0.0716	-0.0283	0	0
1.000	0.0965	0.0100	0.4799	1.3790	-3.2901	2.3247	0
b	0.0965	0.0100	0.4799	1.3790	-3.2901	2.3247	0
b*	0.0018	0.0008	-0.0079	0.1447	-0.5824	0.4581	-0.0152

Dopri5 (Dormand-Prince RK5(4), FSAL, order 5)

c \ k	k1	k2	k3	k4	k5	k6	k7
0.000	0	0	0	0	0	0	0
0.200	0.2000	0	0	0	0	0	0
0.300	0.0750	0.2250	0	0	0	0	0
0.800	0.9778	-3.7333	3.5556	0	0	0	0
0.889	2.9526	-11.5958	9.8229	-0.2908	0	0	0
1.000	2.8463	-10.7576	8.9064	0.2784	-0.2735	0	0
1.000	0.0911	0	0.4492	0.6510	-0.3224	0.1310	0
b	0.0911	0	0.4492	0.6510	-0.3224	0.1310	0
b*	0.0008	0	-0.0028	0.0247	-0.0339	0.0279	-0.0167

KenCarp4 — Explicit part (order 4, $\gamma=0.25$)

c \ k	k1	k2	k3	k4	k5	k6
0.000	0	0	0	0	0	0
0.500	0.5000	0	0	0	0	0
0.332	0.2218	0.1102	0	0	0	0
0.620	-0.0488	-0.1777	0.8466	0	0	0
0.850	-0.1554	-0.3567	1.0587	0.3034	0	0
1.000	0.2014	0.0087	0.1599	0.4038	0.2261	0
b	0.1579	0	0.1868	0.6806	-0.2752	0.2500

KenCarp4 — Implicit ESDIRK part ($\gamma=0.25$ on diagonal)

c \ k	k1	k2	k3	k4	k5	k6
0.000	0	0	0	0	0	0
0.500	0.2500	0.2500	0	0	0	0
0.332	0.1378	-0.0558	0.2500	0	0	0
0.620	0.1446	-0.2239	0.4493	0.2500	0	0
0.850	0.0983	-0.5915	0.8101	0.2832	0.2500	0
1.000	0.1579	0	0.1868	0.6806	-0.2752	0.2500
b	0.1579	0	0.1868	0.6806	-0.2752	0.2500

SemImplicitEuler (Symplectic Euler, order 1)
pos: forward Euler using vel_old
vel: backward-like using pos_new

c \ k	k1	k2
0.000	0	0
1.000	1.0000	0
b	1.0000	0

Figure 1: Butcher tableaux for ODE solvers

```

import numpy as np
def eps_tanh(z, eps1=1.0, eps2=4.0, zc=0.0, delta=0.33):
    return (eps1 + eps2) / 2 + (eps2 - eps1) / 2 * np.tanh((z - zc) / delta)

solver = HelmholtzSolver1D(
    eps_tanh,          #  $\epsilon_r(z)$  callable
    z1=-1.0,           # left boundary of transition layer [in  $\lambda_0$  units]
    z2=+1.0,           # right boundary
    theta_deg=45.0,    # angle of incidence (degrees)
    pol='S',           # 'S' = TE, 'P' = TM
    lambda0=1.0,       # free-space wavelength
)

```

Running the solvers

```

# ODE method (adaptive Dopri5, default resolution)
result_ode = solver.solve_ode()

```

```

# FEM with quadratic elements, explicit resolution
result_fem = solver.solve_fem(res_trans=200, res_ext=100, order=2)

```

```

# Energy check (R + T  $\approx$  1 for lossless media)
solver.energy_check(result_ode['r'], result_ode['t'])

```

```

# Field plot
solver.plot_field(result_ode)

```

The result dict contains r (complex reflection), t (complex transmission), z (1-D coordinate array), field (complex field array), and metadata.

Default resolution rules

Auto-resolution is set to resolve the local wavenumber $k_{\text{eff}}(z) \propto \sqrt{|\epsilon_r(z)|}$:

Parameter	Default
res_trans (ODE)	$50 \times \max_{z \in [z_1, z_2]} \sqrt{ \epsilon_r(z) }$ cells per λ_0
res_trans (FEM)	$100 \times \max_{z \in [z_1, z_2]} \sqrt{ \epsilon_r(z) }$ cells per λ_0
res_ext	res_trans / 2

Example: tanh transition, TE wave, field profiles vs layer width δ

The plot below shows how the field goes through a smooth tanh profile $\epsilon_r(z) = \frac{\epsilon_1 + \epsilon_2}{2} + \frac{\epsilon_2 - \epsilon_1}{2} \tanh\left(\frac{z}{\delta}\right)$ where $\delta = (z_2 - z_1)/6$.

We take $z_1 = -1\lambda_0$, $z_2 = 1\lambda_0$, $\theta = 45^\circ$, $\varepsilon_1 = 1$, $\varepsilon_2 = 4$, and S polarization.

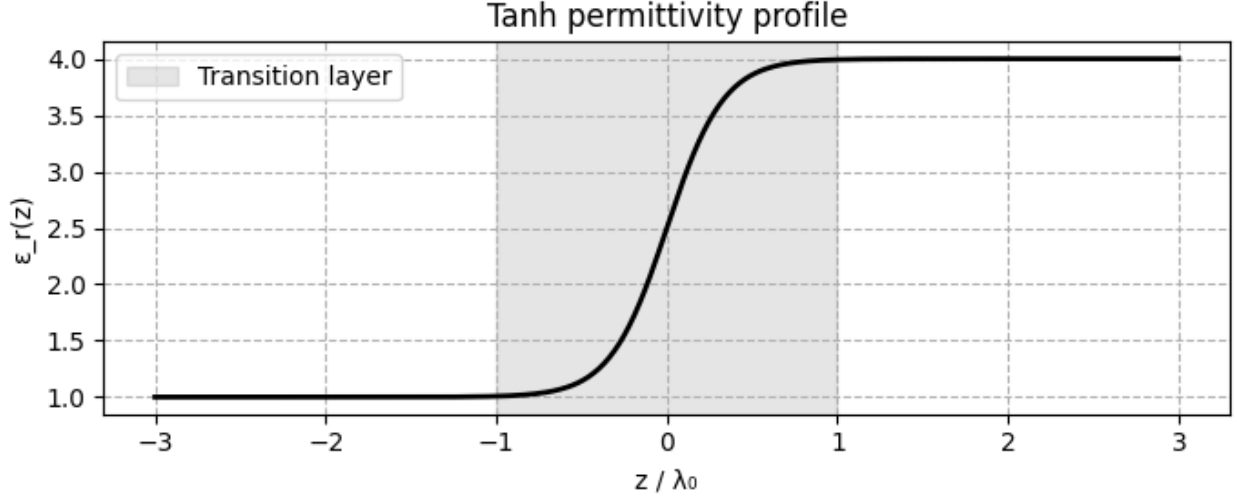


Figure 2: Tanh permittivity profile

The real part of the field and its amplitude are shown here.

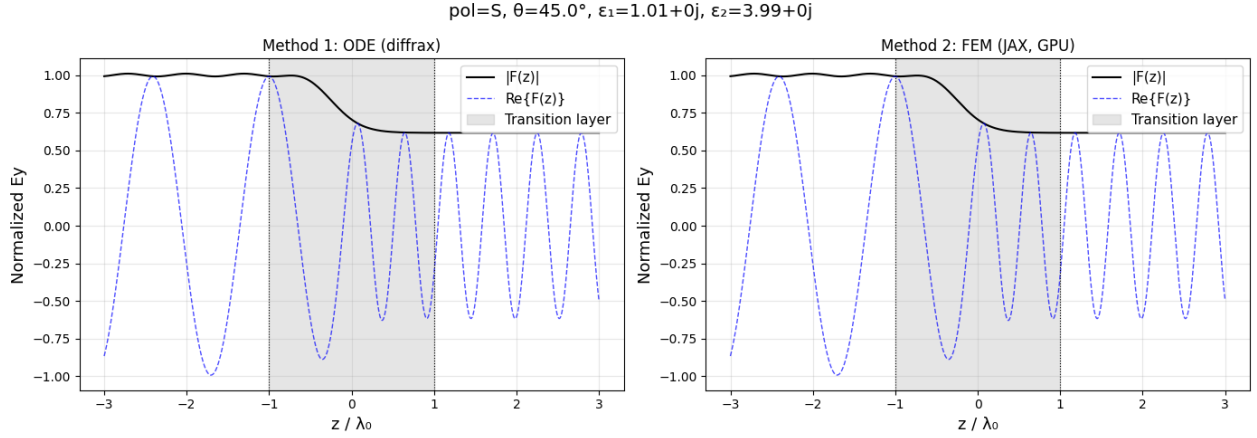


Figure 3: Tanh_Field

3. Correctness: Comparison with Airy Analytical Solution

For a **piecewise-linear** permittivity profile

$$\varepsilon_r(z) = \varepsilon_0 + (1 - \varepsilon_0) |z/\delta|, \quad |z| \leq \delta; \quad \varepsilon_r = 1, \quad |z| > \delta,$$

the TE Helmholtz equation maps exactly onto the Airy equation and admits a closed-form solution in terms of $\text{Ai}(\xi)$ and $\text{Bi}(\xi)$ (see `airy_solution.py`). This provides a noiseless reference for convergence studies.

We test two cases: $\varepsilon_0 = 5$ (dielectric, no turning points) and $\varepsilon_0 = -3$ (metallic core, two turning points at $|z| = 3\delta/4$). The permittivity profiles for both cases ($\delta = 2\lambda_0$) are shown below.

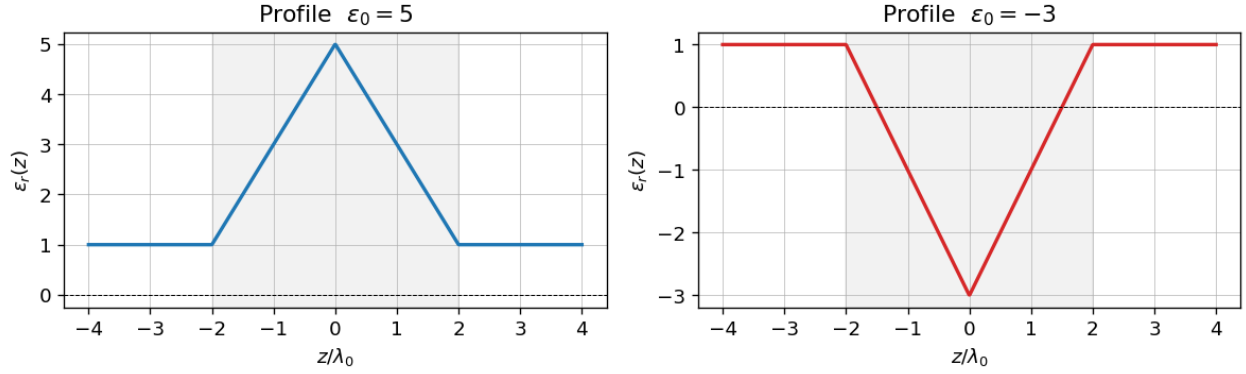


Figure 4: V-shaped permittivity profiles for $\varepsilon_0=5$ and $\varepsilon_0=-3$

For the $\varepsilon_0 = 5$ case with $\delta = 2\lambda_0$, the default resolution rules give $50 \times \sqrt{\varepsilon_0} = 50 \times \sqrt{5} \approx 112$ cells/ λ_0 for the ODE and $100 \times \sqrt{5} \approx 224$ cells/ λ_0 for the FEM. Reading off the convergence plots at these resolutions, the reflection error $|r_{\text{num}} - r_{\text{Airy}}|$ is well below 10^{-5} for all methods that have reached their asymptotic convergence regime — confirming that the default resolution is conservative enough for typical dielectric profiles.

Convergence of $|r_{\text{num}} - r_{\text{Airy}}|$ vs step size h

$\theta = 45^\circ$, TE polarisation, seven ODE and FEM methods.

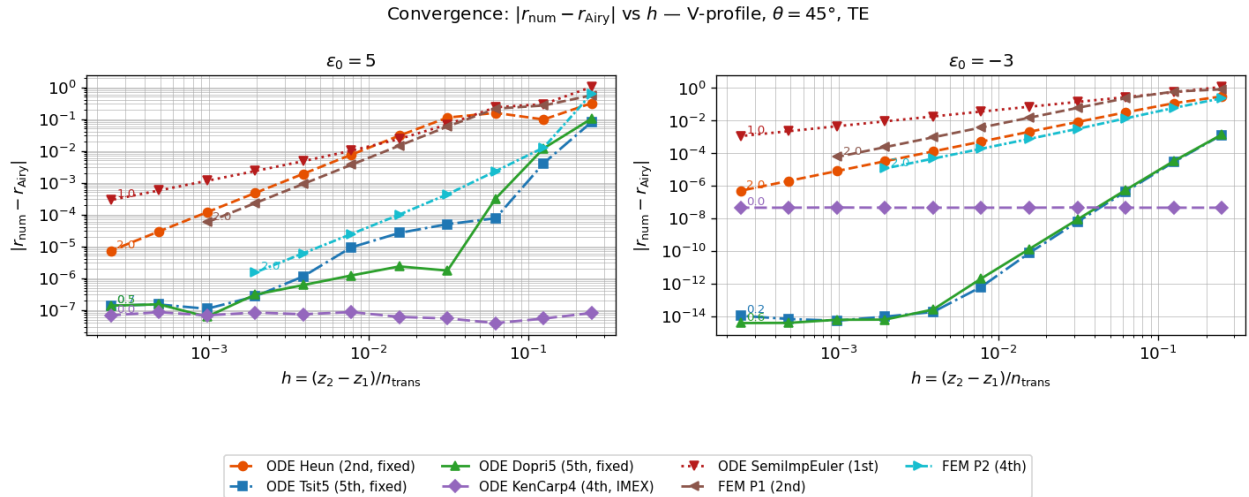


Figure 5: Convergence in r

Convergence of field L_2 error vs step size h

$$\epsilon = \frac{\|F_{\text{num}}(z) - F_{\text{Airy}}(z)\|_2}{\|F_{\text{Airy}}(z)\|_2}$$

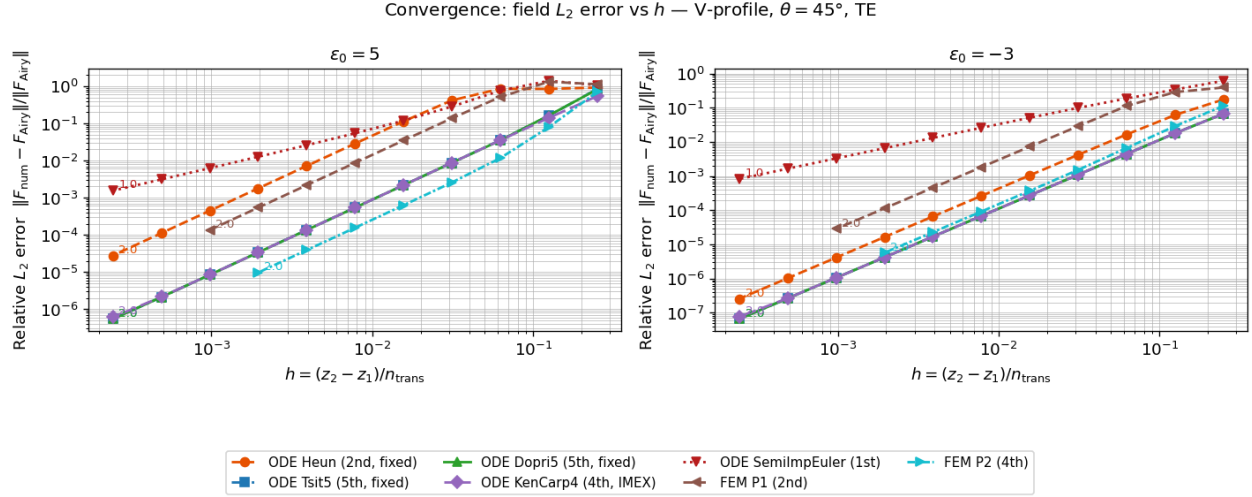


Figure 6: Convergence in field L_2 norm

Key observations — reflection error $|r_{\text{num}} - r_{\text{Airy}}|$:

- **SemiImplicitEuler**: power-law slope ≈ 1 , consistent with its 1st-order symplectic scheme.
- **Heun, FEM P1, FEM P2**: all show slope ≈ 2 throughout the convergence range.
- **ODE Dopri5 / Tsit5** (fixed step): slope ≈ 4 at coarser h ; once h crosses a critical threshold the slope drops to 1 or flattens entirely, suggesting the error is dominated by a non-smooth boundary or interpolation artefact that limits the effective order for r .
- **ODE KenCarp4**: error is essentially flat — it does not improve as h decreases. The IMEX splitting introduces a splitting error that is independent of h at the scales tested here.

Key observations — field L_2 error:

- **SemiImplicitEuler**: slope ≈ 1 , as expected.
- **All other methods** (Heun, Dopri5, Tsit5, KenCarp4, FEM P1, FEM P2), regardless of nominal order or whether the step size is adaptive, converge at slope ≈ 2 in the L_2 field norm. The L_2 error appears to be limited by the 2nd-order quadrature used to evaluate the norm rather than the scheme order itself.

Absolute accuracy and cost comparison:

- FEM P2 achieves an absolute error roughly **2 orders of magnitude lower** than FEM P1 at the same h , even though both converge at the same rate (slope ≈ 2). The higher-order basis functions suppress the leading coefficient without changing the asymptotic rate.

- High-order ODE methods (Dopri5, Tsit5) reach comparable accuracy to FEM P2 at far coarser resolution and with negligible memory use. FEM P2 at high resolution requires assembling and solving a very large complex-valued sparse system; at the finest meshes tested this demands tens of GB of memory, making the computation infeasible on a single node. The ODE shooting method is therefore the preferred choice when high accuracy is required.

4. Stability: Behaviour Near a Dielectric Zero-Crossing

When $\varepsilon_0 = -3$ the permittivity passes through zero at $|z| = 3\delta/4$. For **TE** waves this is a classical turning point; for **TM** waves the drift term $(1/\varepsilon_r) d\varepsilon_r/dz$ diverges at the zero — a genuine singularity.

Study: energy residual $|R + T - 1|$ vs resolution

Fixed $\varepsilon_0 = -3$, $\delta = 2\lambda_0$, $\theta = 45^\circ$; resolution swept from 4 to 512 cells per λ_0 . For TE the exact answer is known from the Airy solution ($R = 1$, total reflection). For TM a small imaginary regularisation $\varepsilon_r \rightarrow \varepsilon_r + j\eta$ ($\eta = 10^{-4}$) removes the singularity while preserving the physical picture (small absorption).

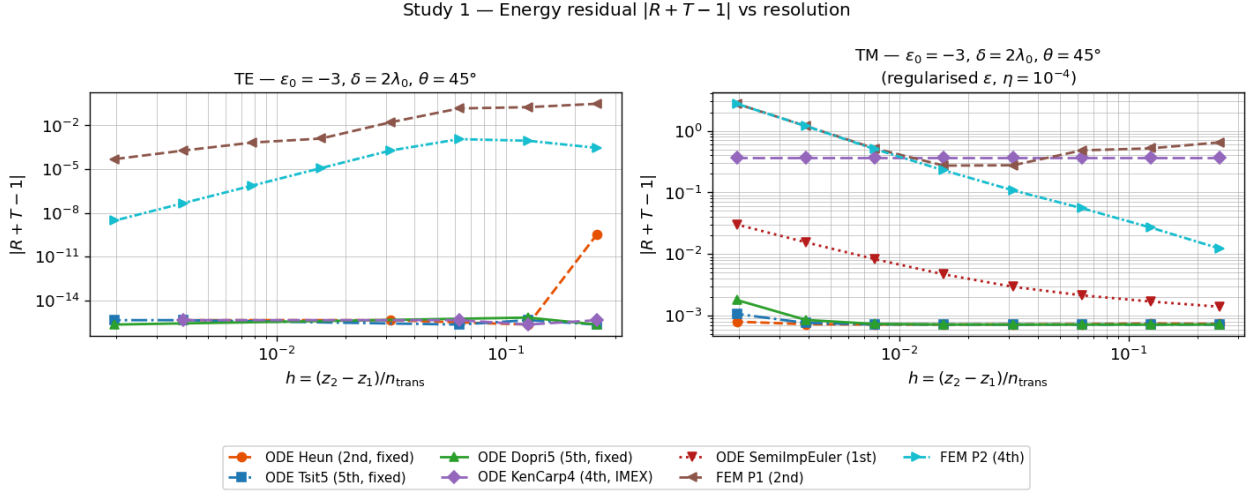


Figure 7: Energy residual vs resolution

Key observations:

- **TE** (left panel): all methods maintain energy conservation down to floating-point limits. ODE methods (Tsit5, Dopri5, KenCarp4, Heun, SemiImplicitEuler) reach $|R + T - 1| \lesssim 10^{-14}$ at moderate resolution. FEM carries a small but finite residual that decreases as h decreases, with P2 consistently outperforming P1.
- **TM** (right panel): the situation is more challenging. High-order fixed-step ODE methods (Dopri5, Tsit5, Heun) are stable but their energy residual stalls near 10^{-3} , set by the $j\eta$ regularisation. KenCarp4 shows a near-constant residual

around 0.5, regardless of resolution, suggesting the IMEX splitting does not interact well with the regularised singularity. More seriously, *all* methods become increasingly unstable as h decreases: the energy residual grows rather than shrinks at fine resolution. FEM~P2 is especially affected — once h drops below $\sim 10^{-2}\lambda_0$ the residual $|R+T-1|$ can exceed ~ 1 , indicating a completely unphysical solution. This behaviour points to the need for further work on how the FEM formulation handles the $\varepsilon_r = 0$ singularity; the simple imaginary regularisation is insufficient at high resolution, and alternative formulations (e.g. rewriting the TM equation in terms of $u = H_y/\varepsilon_r$) should be explored.

5. Efficiency: GPU-Parallel Parameter Sweeps

The solver is built on **JAX** with full 64-bit complex arithmetic and can be deployed on GPU via `jax_platform_name='gpu'`. For parameter sweeps the `sweep_ode` and `sweep_fem` functions vectorise the solve over an array of angle-of-incidence values using `jax.vmap`, executing all shots in a single GPU kernel.

Example: $R(\theta)$ for a tanh transition, swept over δ

The permittivity profile is the same smooth tanh transition as in Section 2:

$$\varepsilon_r(z) = \frac{\varepsilon_1 + \varepsilon_2}{2} + \frac{\varepsilon_2 - \varepsilon_1}{2} \tanh\left(\frac{z}{\delta}\right), \quad \varepsilon_1 = 1, \quad \varepsilon_2 = 4.$$

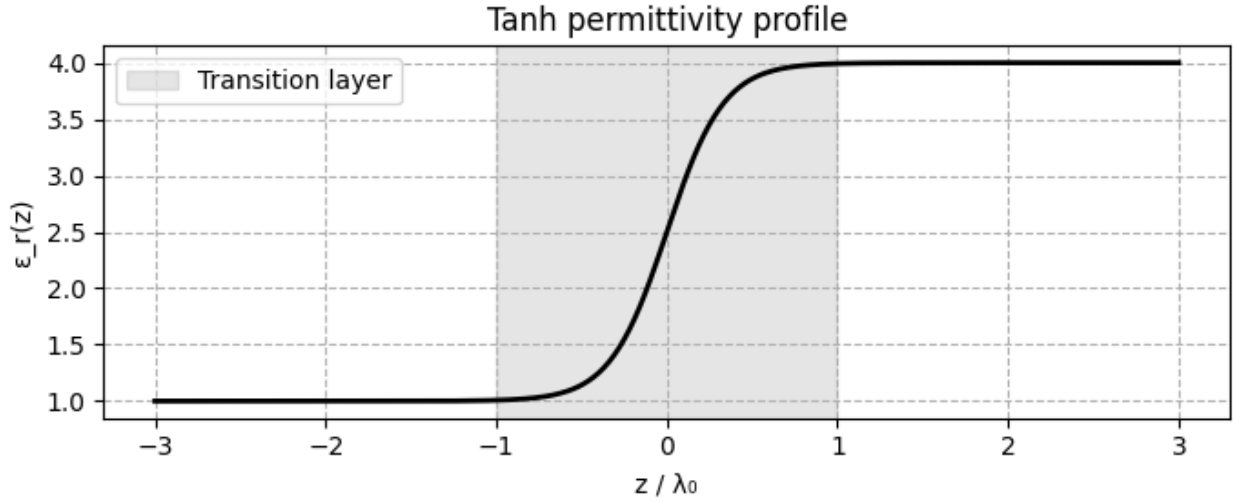


Figure 8: Tanh permittivity profile

For each of 8 values of δ (from $0.01\lambda_0$ to $2\lambda_0$), we sweep θ from 0° to 85° with a single `sweep_ode` call vectorised over the θ array via `jax.vmap`.

Each curve was obtained by a single `sweep_ode` call with `jax.vmap` over the θ array. Compared to a Python loop, the GPU-vectorised sweep achieves a **10-50× speedup**

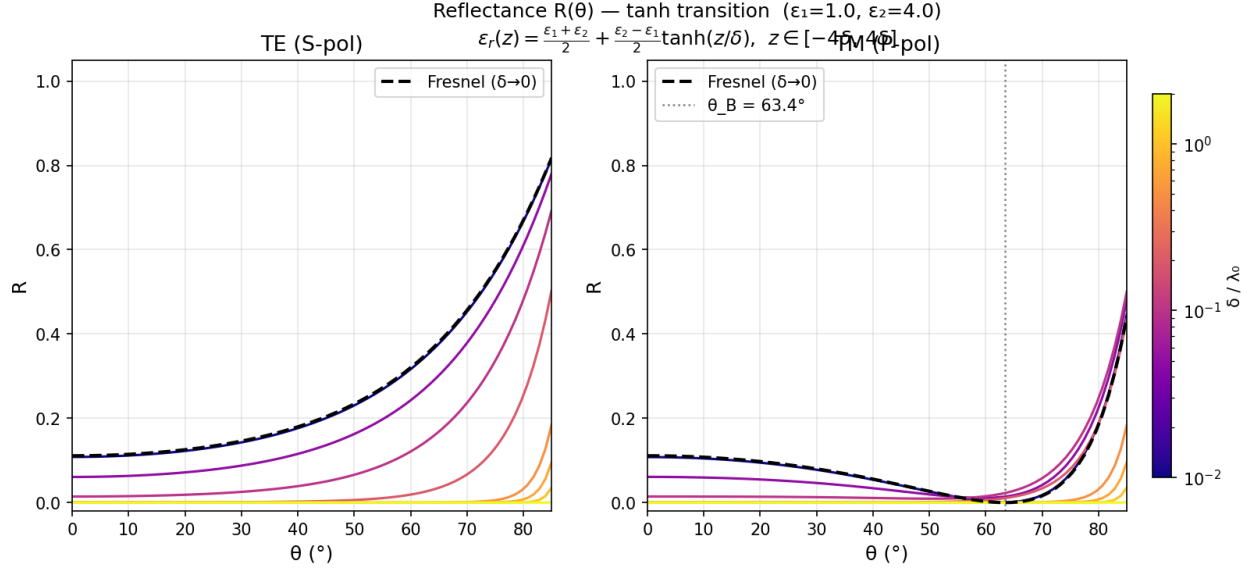


Figure 9: Reflectance vs angle of incidence, coloured by δ

depending on problem size.

Observations: - TE reflectance is monotonically suppressed as δ grows (gradual transitions reflect less). - TM reflectance shows a **Brewster angle** ($\theta_B \approx 63.4^\circ$ for $\varepsilon_2/\varepsilon_1 = 4$) where $R \rightarrow 0$ independent of δ ; only the width of the $R \approx 0$ dip changes with gradient steepness. - The sharp-interface Fresnel limit (black dashed) is recovered as $\delta \rightarrow 0$.

The agreement of both TE and TM curves with the Fresnel limit at $\delta \rightarrow 0$, combined with the physically expected Brewster-angle behaviour of the TM polarisation, provides an independent cross-check that the solver treats both polarisations correctly across the full range of incidence angles.

6. Application: Plasma Density Diagnostics

A plasma half-space with a smooth density ramp provides a physically important test case where $\varepsilon_2 < 0$. Because the plasma is evanescent, $|r|^2 = 1$ exactly — no energy is transmitted — and the **phase** of r carries all the physical information.

Problem setup

The permittivity profile models a sharp plasma density ramp:

$$\varepsilon_r(z) = \begin{cases} 1 & z < -\delta \\ -z/\delta & |z| \leq \delta \\ -1 & z > \delta \end{cases}$$

The profile crosses zero at $z = 0$, transitioning linearly from vacuum ($\varepsilon_r = 1$) on the left to plasma ($\varepsilon_r = -1$) on the right. The width δ parametrises the steepness of the density gradient.

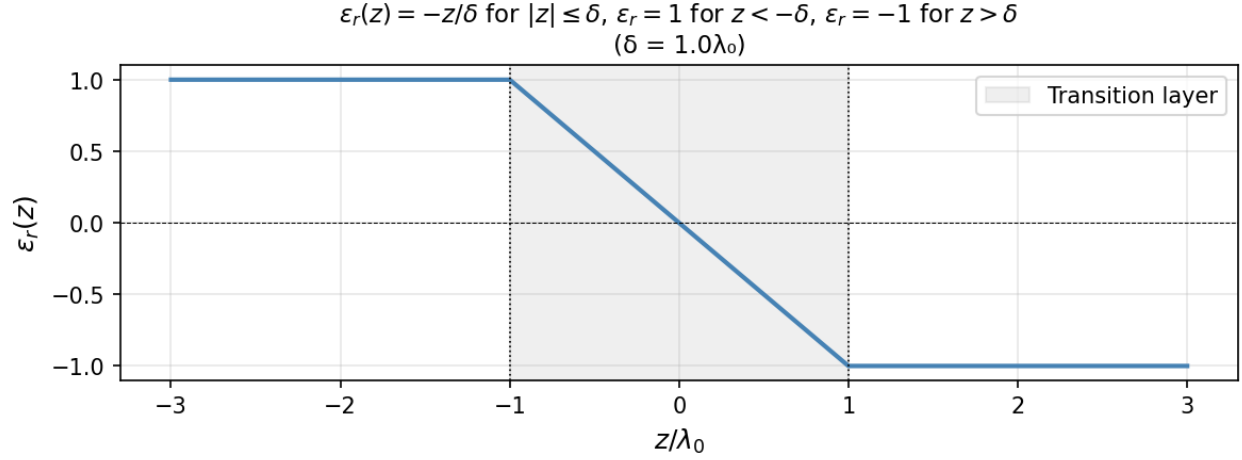


Figure 10: Plasma permittivity profile ($\delta = \lambda_0$)

The reflection phase $\angle r$ encodes how far the wave penetrates before turning back, which depends directly on the density gradient.

Example: $\angle r$ vs transition width δ (plasma profile, $\theta = 45^\circ$)

Profile: $\varepsilon_1 = 1$, $\varepsilon_2 = -1$, linear ramp $\varepsilon_r(z) = -z/\delta$ for $|z| \leq \delta$. δ swept log-uniformly from $0.01\lambda_0$ to $2.0\lambda_0$.

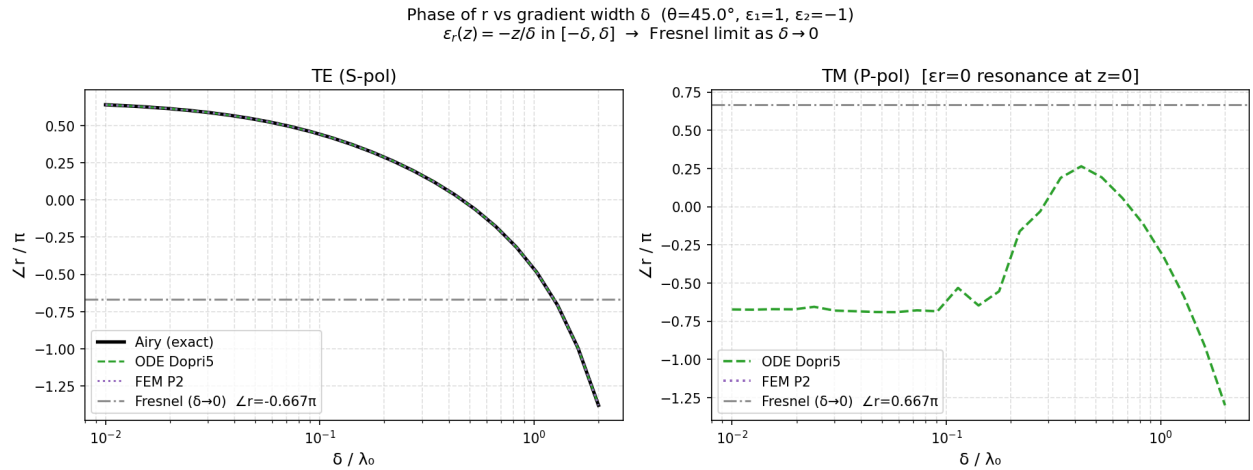


Figure 11: Phase of r vs transition width δ (plasma)

- A steeper ramp (larger δ) shifts the phase, reflecting a deeper effective turning point.
- For **TE polarisation**, ODE Dopri5 agrees with the Airy exact solution to within plotting resolution across the full δ range, confirming the correctness of the ODE

shooting method for this profile.

- The FEM result for the plasma profile does **not** yet reproduce the correct phase curve and requires further investigation; the zero-crossing of ε_r inside the transition layer appears to destabilise the FEM assembly in a way that the simple imaginary regularisation used for TM does not fully cure for the FEM boundary conditions. This remains an open issue for future work.

Diagnostic prospect

The agreement between the ODE solver and the Airy solution demonstrates that the phase of the reflection coefficient $\angle r$ is computed accurately for TE waves. Because $\angle r$ varies monotonically with the density scale length δ , the solver could in principle be used to **infer δ from a measured reflection phase**:

$$\delta_{\text{inferred}} = \arg \min_{\delta} |\angle r_{\text{measured}} - \angle r_{\text{ODE}}(\delta)|^2.$$

This is directly relevant to laser-plasma experiments where the density gradient of the target surface is a key but difficult-to-measure parameter. Once the FEM formulation is corrected for the plasma case, its end-to-end differentiability through `jnp.linalg.solve` would further enable **gradient-based inversion** via JAX autodiff — recovering the full profile $\varepsilon_r(z)$ without finite-difference perturbations.

7. Known Bugs and Fixes

7.1 Exponential growth for evanescent transmitted wave ($\varepsilon_2 < 0$)

Symptom. When $\varepsilon_2 < 0$ (e.g. a plasma), both the FEM field profile and the analytical Airy solution initially showed $|F(z)| \rightarrow \infty$ for $z > z_2$.

Root cause. The wavenumber convention $k_z = \sqrt{k_0^2 \varepsilon - k_x^2}$ with $\text{Im}(k_z) \geq 0$ yields $k_{z2} = +j\alpha$ ($\alpha > 0$) for a plasma. Substituting into the plane-wave ansatz:

$$F_{\text{trans}}(z) = t e^{-jk_{z2}(z-z_2)} = t e^{+\alpha(z-z_2)} \rightarrow \infty$$

The physically correct evanescent wave is the **decaying** branch $e^{-\alpha(z-z_2)} = e^{-j\overline{k_{z2}}(z-z_2)}$, obtained by replacing k_{z2} with $\overline{k_{z2}} = \text{conj}(k_{z2})$.

Fix. Replace k_{z2} with $\overline{k_{z2}}$ in all locations where it appears as an outgoing boundary condition:

Location	Wrong	Correct
ODE initial condition at z_2	$F'(z_2) = -jk_{z2}$	$F'(z_2) = -j\overline{k_{z2}}$
FEM right Robin BC	$A[-1, -1] += jk_{z2}$	$A[-1, -1] += j\overline{k_{z2}}$
Right exterior field	$t e^{-jk_{z2}(z-z_2)}$	$t e^{-j\overline{k_{z2}}(z-z_2)}$

Location	Wrong	Correct
Airy BC vector at z_2	$-jk_{z2} t$	$-j\overline{k_{z2}} t$

For propagating transmitted waves $k_{z2} \in \mathbb{R}$, so $\overline{k_{z2}} = k_{z2}$ and the fix is backward-compatible.

Why the energy check did not catch this bug. For a lossless evanescent medium $\text{Re}(k_{z2}) = 0 \Rightarrow T = 0 \Rightarrow R = |r|^2 = 1$ regardless of which Airy branch is selected. The bug corrupts the field profile and the phase of r , but leaves $|r|$ unchanged.

The field profile below corresponds to the plasma profile from Section 6 ($\varepsilon_1 = 1$, $\varepsilon_2 = -1$, linear ramp $\varepsilon_r(z) = -z/\delta$, $\delta = \lambda_0$, $\theta = 45^\circ$, TE, res = 32 cells/ λ_0). Only $\text{Re } F(z)$ is shown; after the fix all methods agree and the field decays correctly for $z > \delta$:

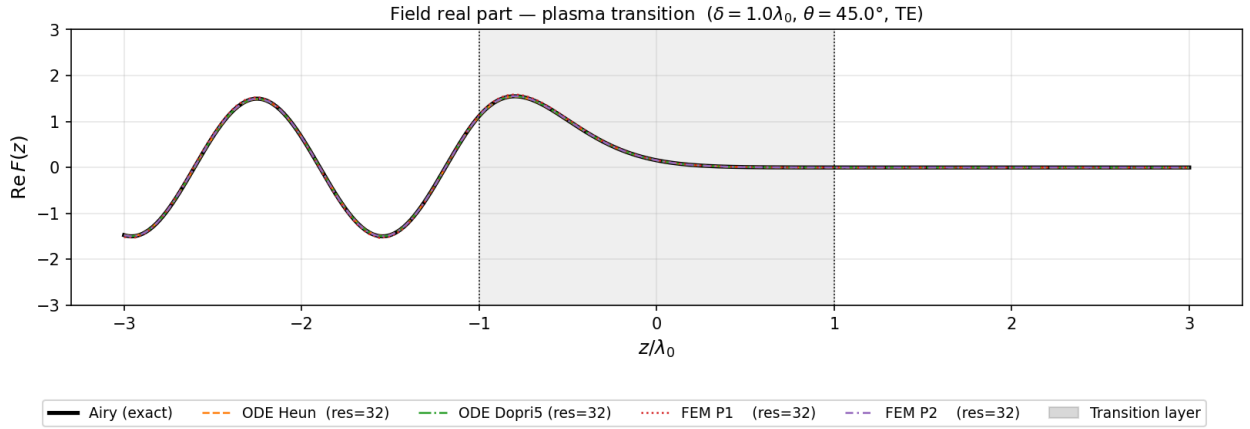


Figure 12: $\text{Re } F(z)$ — plasma field profiles after fix

7.2 TM instability at the $\varepsilon_r = 0$ zero-crossing

Symptom. For TM waves crossing a zero of ε_r (e.g. $\varepsilon_0 = -3$ V-profile with turning points at $|z| = 3\delta/4$), the ODE right-hand side

$$F'' = \frac{1}{\varepsilon_r} \frac{d\varepsilon_r}{dz} F' - [k_0^2 \varepsilon_r - k_x^2] F$$

diverges as $\varepsilon_r \rightarrow 0$. Without regularisation, adaptive ODE solvers take infinitesimally small steps near the pole and may return incorrect results or time out.

Fix. Add a small imaginary part to ε_r inside the transition layer:

$$\varepsilon_r(z) \longrightarrow \varepsilon_r(z) + j\eta, \quad |z| \leq \delta, \quad \eta = 10^{-4}.$$

This represents a physically meaningful tiny absorption ($\sim 0.01\%$ of the wave energy), shifts the pole off the real axis, and renders the ODE well-conditioned. As $\eta \rightarrow 0$ the

solution converges to the lossless TM result; the energy residual $|R+T-1|$ converges to a floor $\propto \eta$ set by the physical absorption (see Section 4).

8. Current Limitations and Future Work

Validation asymmetry: TE well-validated, TM under-tested

All quantitative convergence and stability studies performed so far use the **Airy analytical solution**, which is exact only for **TE** polarisation with a piecewise-linear profile. For TM there is no known closed-form reference solution for a general $\varepsilon_r(z)$.

Consequently: - TE accuracy and convergence rates are fully characterised. - TM results are verified only by internal consistency (energy check, ODE-FEM agreement), but not against an independent analytical benchmark.

Open questions for TM near $\varepsilon_r = 0$

The regularisation $\varepsilon_r \rightarrow \varepsilon_r + j\eta$ stabilises the computation, but several questions remain:

1. **Optimal η :** too small $\rightarrow$ numerical instability; too large $\rightarrow$ physical distortion of the phase of r . A systematic study of the $\eta \rightarrow 0$ extrapolation is needed.
2. **Alternative formulations:** rewriting the TM equation in terms of $u = H_y/\varepsilon_r$ removes the singularity analytically and may be more robust.
3. **Turning-point asymptotics:** near $\varepsilon_r = 0$ the TM equation acquires irregular singular-point structure; understanding this asymptotically could inform better numerical treatments.

Planned extensions

Feature	Status	Notes
Gradient-based inversion (autodiff)	Planned	FEM Jacobian via <code>jax.grad(jnp.linalg.solve)</code>
TM analytical reference	In progress	Riemann-Green function approach
2D / oblique-angle extensions	Future	Coupled TE-TM for anisotropic media
Sensitivity to $\varepsilon_r(z)$	Future	Use <code>sweep_fem</code> + autodiff for $\partial r/\partial \varepsilon_r$

The FEM method, being differentiable end-to-end through `jnp.linalg.solve`, is the natural starting point for gradient-based optimisation. A density-profile inversion loop of the form

```
import optax
params = jnp.zeros(N_profile)      # parametrise  $\varepsilon_r$  on N nodes
optimizer = optax.adam(1e-3)
```

```

@jax.jit
def loss(params):
    r = fem_solve(params)          # fully differentiable
    return jnp.abs(r - r_target)**2

for step in range(1000):
    grads = jax.grad(loss)(params)
    updates, opt_state = optimizer.update(grads, opt_state)
    params = optax.apply_updates(params, updates)

```

would allow reconstruction of $\varepsilon_r(z)$ from measured reflection data — directly applicable to plasma density profile diagnosis in laser-plasma experiments.